ETH zürich

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

NextRAN-AI – 03/07/2026

Integrated Systems Laboratory (ETH Zürich)

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PULP Platform
Open Source Hardware, the way it should be!

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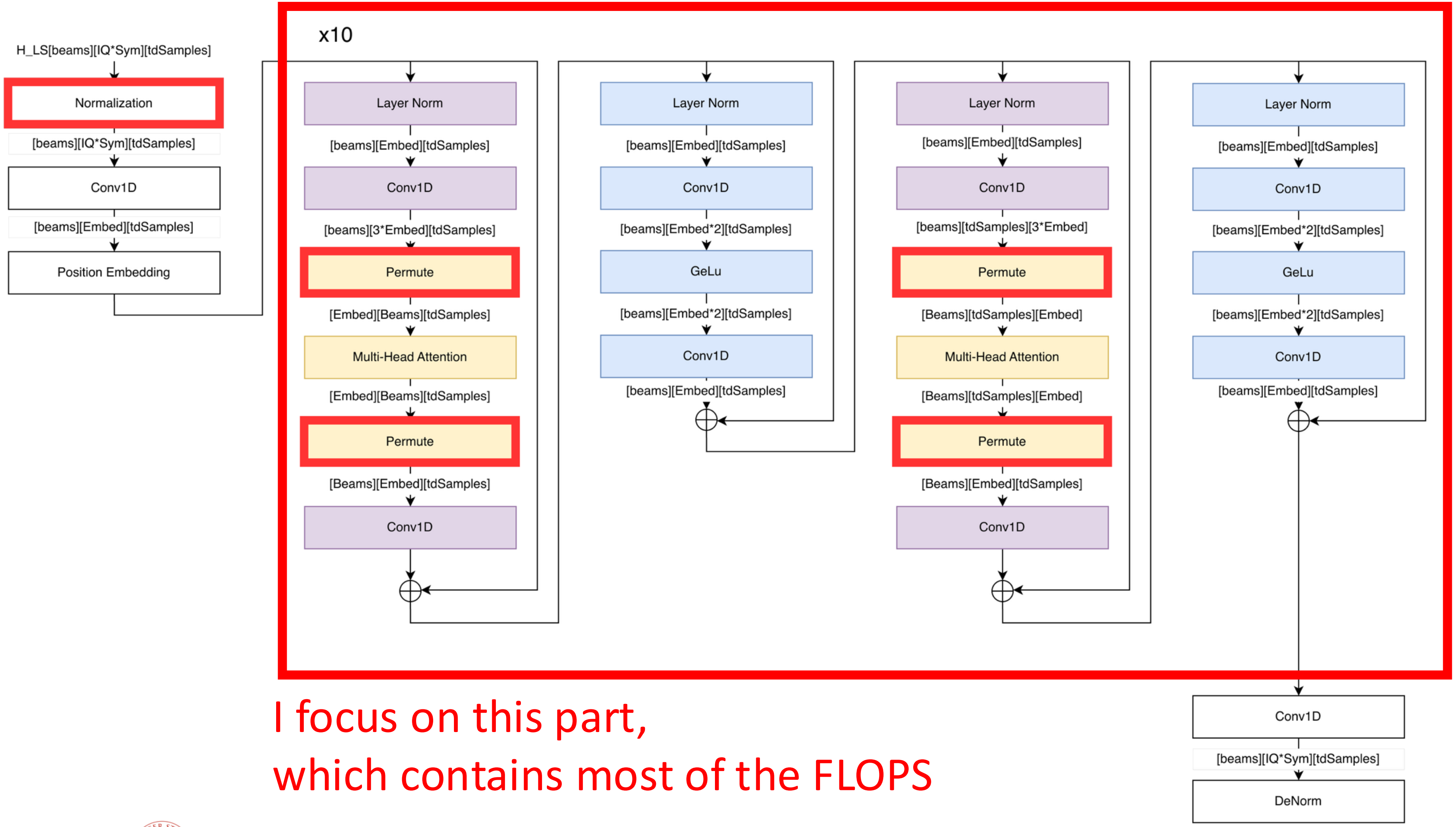
L1 transformer

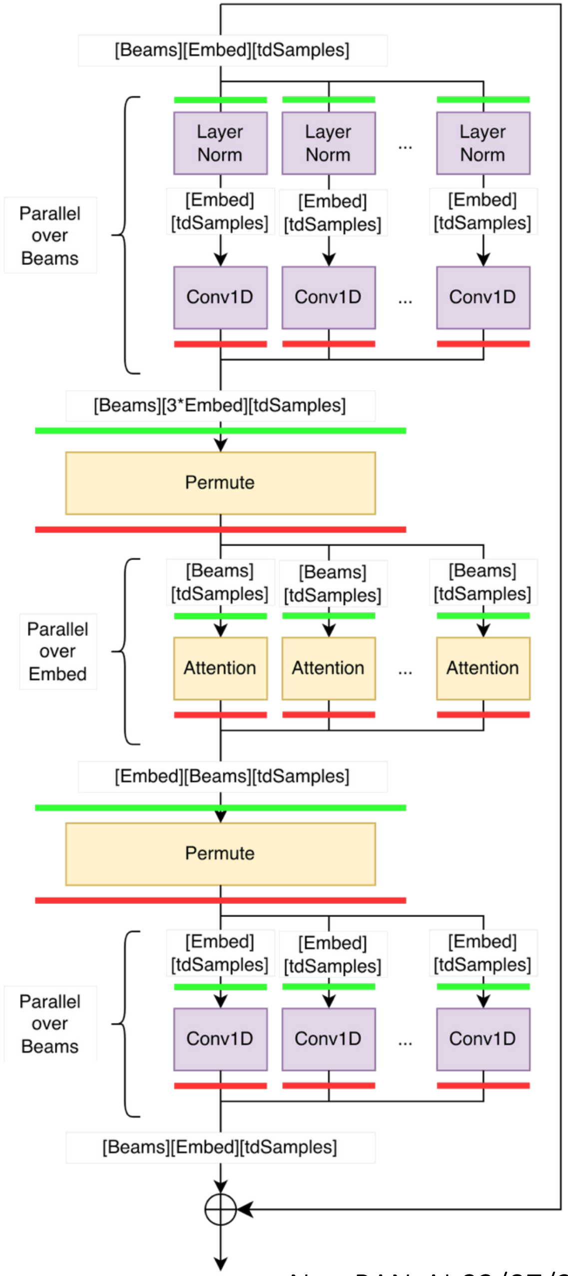


I focus on this part,
which contains most of the FLOPS

Example: Attention-td

(the parallelization of FFN and Attention-Embed will work similarly)

- Parallelize blocks over the external dimension over different clusters → a parallel block turns into parallel serial blocks
- Every serial block is surrounded by loads from L2 and stores to L2



DMA transfers + transposition

- Compute Conv1D, Attention, … in L1
- Store in L2 using the DMA, with the correct shape required by the next computation round

Example: from [Beams][Embed][tdSamples] to [Embed][Beams][tdSamples] requires Beams*Embed DMA transfers of size tdSamples

Embed

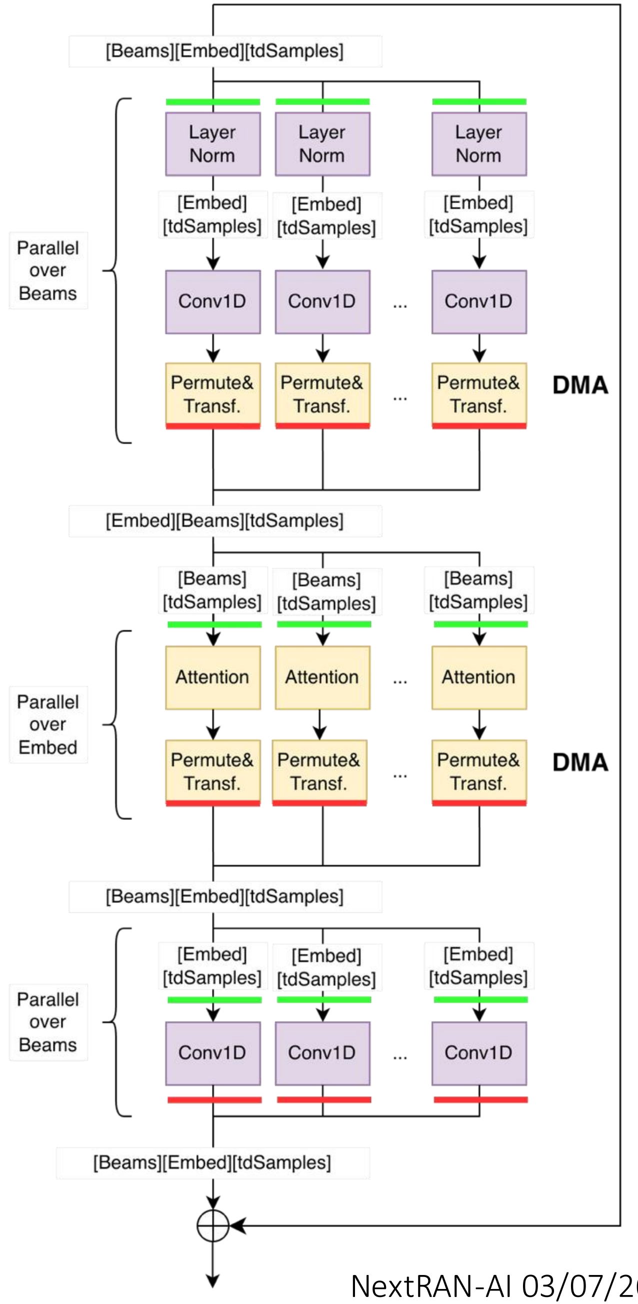
Beams

[tdSamples][tdSamples][tdSamples]
[tdSamples][tdSamples][tdSamples]

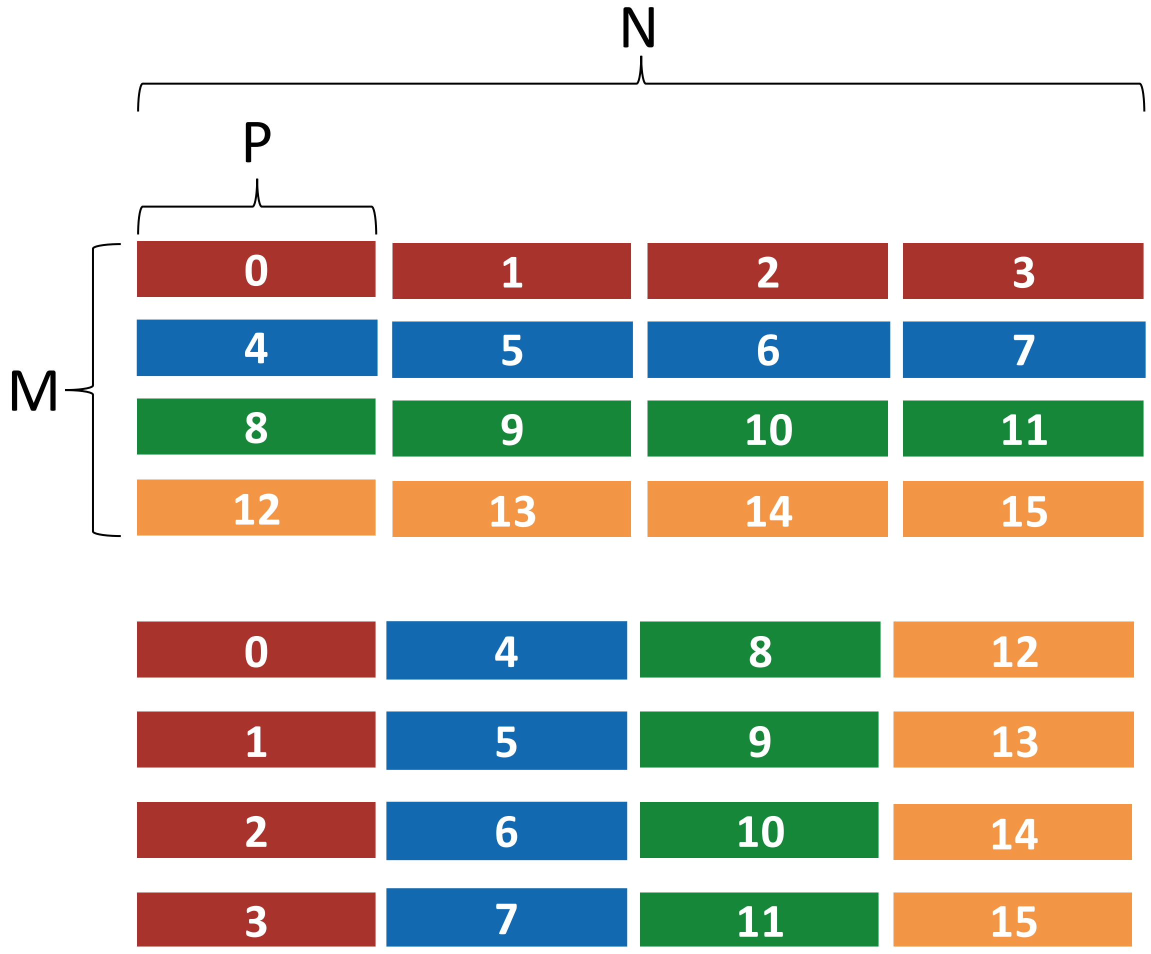
[tdSamples][tdSamples]
[tdSamples][tdSamples]
[tdSamples][tdSamples]

Embed

Beams

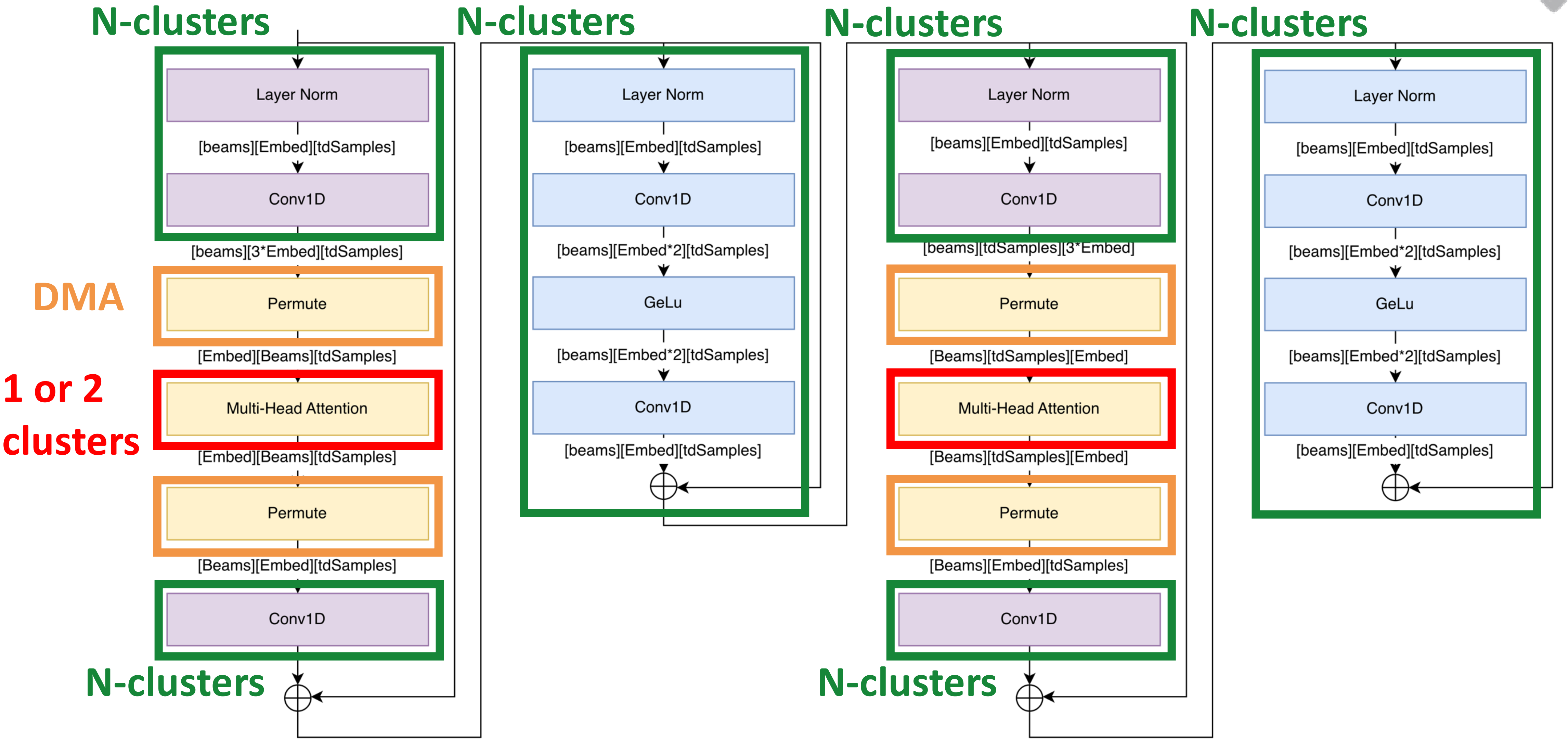


L1 Transformer Permute

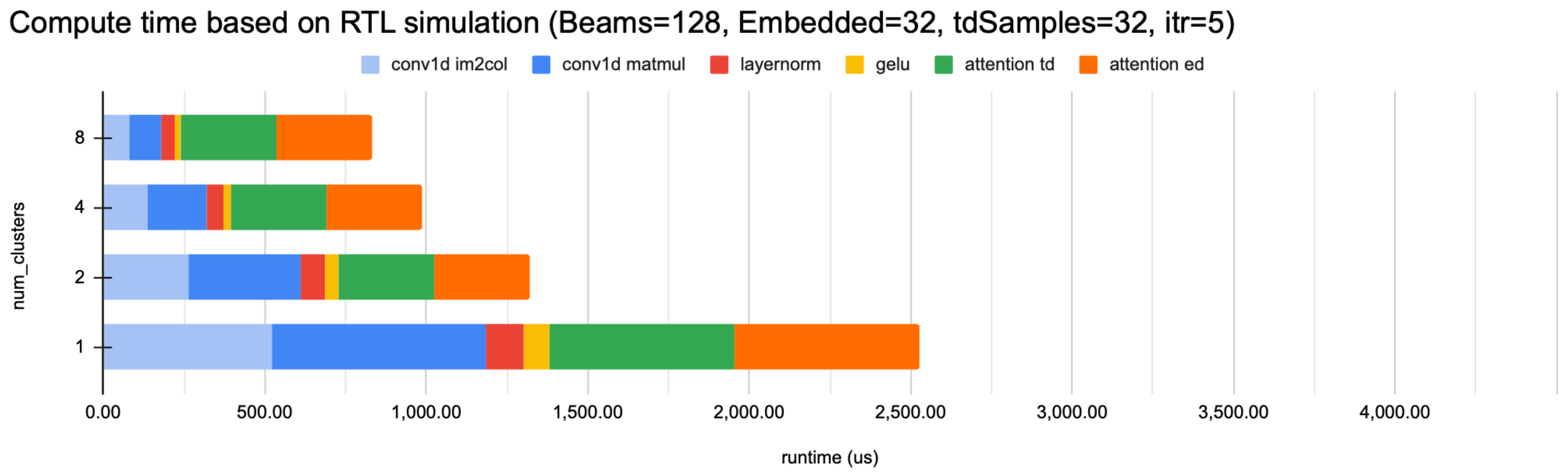


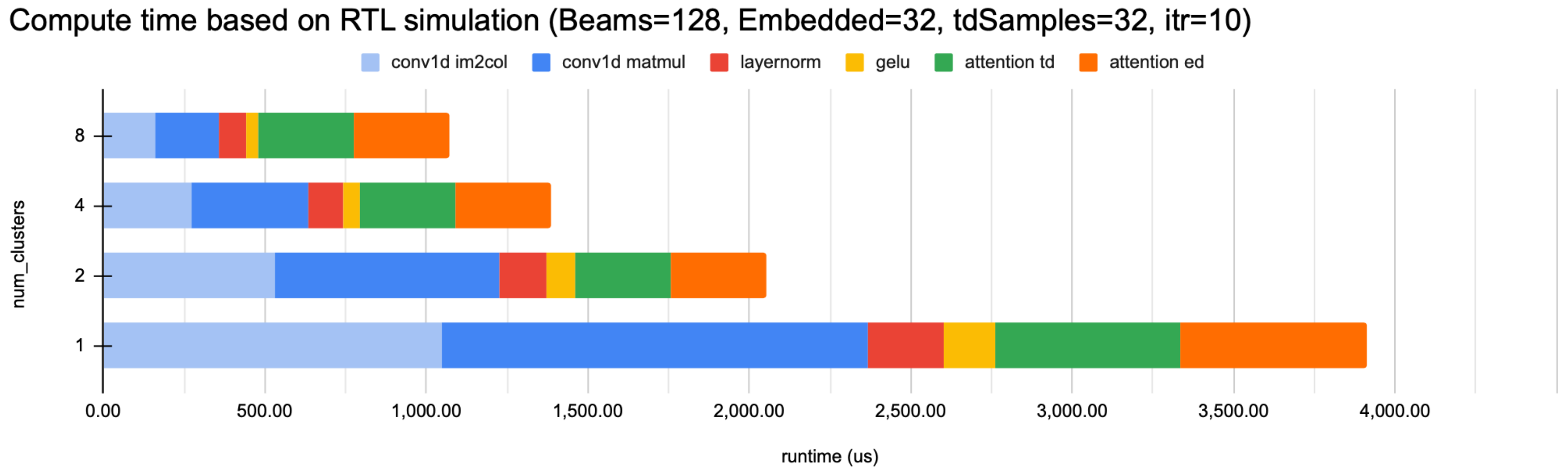

```
/* Permutation from [M][N][P] to [N][M][P] */

uint EL_WIDTH = sizeof(__fp16);
for (uint i = 0; i < M; i++) {
        dma_2dtransfer(
                P * EL_WIDTH,     // num bytes
                M * P * EL_WIDTH, // stride out
                P * EL_WIDTH,     // stride in
                N);               // num copies
}
```

L1 transformer



Runtime in RTL simulation (900MHz)

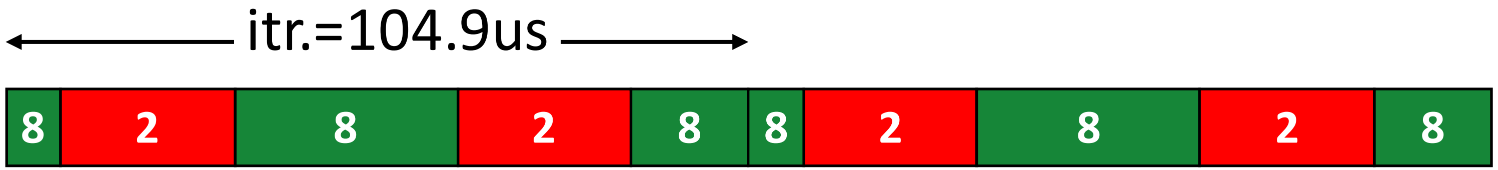




Pipelining parallelism still applied for higher throughput

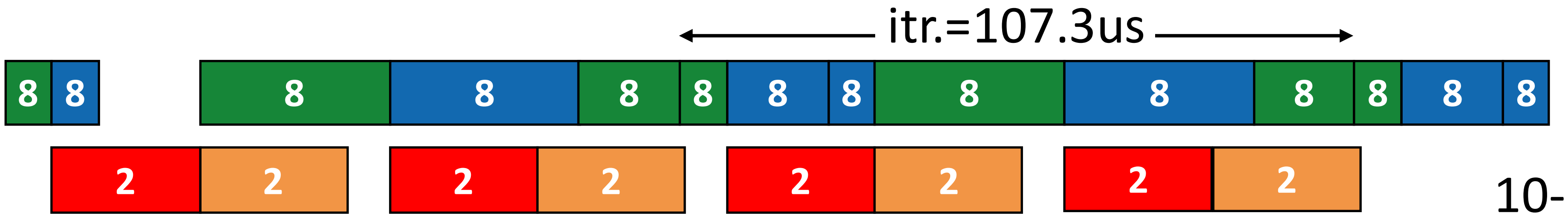
Sequential execution for a group of 8-clusters.

Attention (red) is executed on 2-clusters, 6-clusters are idle for 56% of the time.



Pipelined execution of 2 models on 10-clusters.

2-clusters are idle for 10% of the time.



Almost 2x throughput

Next steps and conclusion of the project

- Complete model evaluation including data movement on GVSoC
- Optimizations on the tensor unit and TCDM connection
- Clean up, upstream of contribution to MemPool main for the final release